

SMART RESUME ANALYZER | SYSTEM ARCHITECTURE

DIAGRAM

Modular Recruitment Analytics & Deterministic ATS Screening Engine

PYTHON FLASK
3.1

GUNICORN 23.0
WSGI

RENDER CLOUD
NATIVE

V2.1
ARCHITECTURE



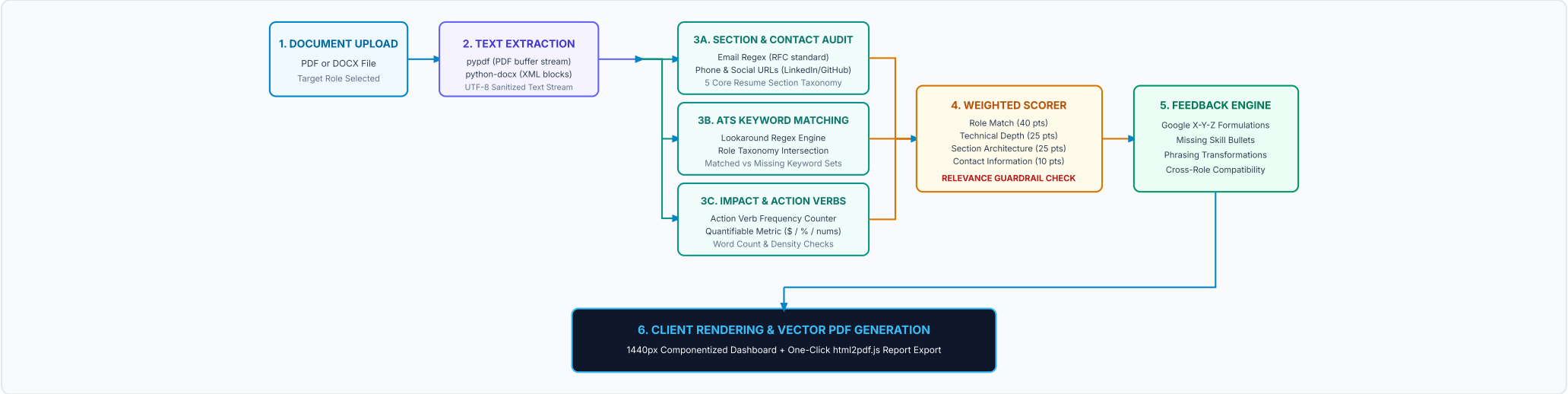
SMART RESUME ANALYZER | DATA FLOW & PIPELINE SPECIFICATION

Step-by-Step Processing Lifecycle, Multi-Factor Scoring Formula & Guardrails

DETERMINISTIC FLOW

RELEVANCE GUARDRAILS

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MATHEMATICAL COMPATIBILITY SCORING MODEL

Score = (RoleMatch × 0.40) + (TechDepth × 0.25) + (SectionArch × 0.25) + (Contact × 0.10)

Anti-Inflation Guardrails:

- If Role Match = 0% → Score capped at ≤ 25/100
- If Role Match < 25% → Score capped at ≤ 40/100
- If Role Match < 50% → Score capped at ≤ 55/100

1. Lookaround Boundary Regex Engine (Eliminating Programming Language Token Bleed)

core/ats_scorer.py

Standard regex word boundaries (\b) fail on programming languages containing symbols. Our engine implements custom lookaround patterns:

```
pattern = rf"(?![a-zA-Z0-9]){re.escape(keyword)}(?![a-zA-Z0-9])"
```

Key ATS Guarantees: Prevents C from matching C++, C#, or CSS; prevents Java from matching JavaScript; accurately captures punctuated tokens like Node.js, ASP.NET, and CI/CD.

2. Empirical Validation: Eliminating Score Inflation (Full Test Results)

TEST CANDIDATE PROFILE	TARGET JOB ROLE	TRADITIONAL SCREENER SCORE	OUR ENGINE GUARDRAILED SCORE	EVALUATION & RECRUITER OUTCOME
Personal Trainer Resume	Web Developer	65 / 100 (False Pass)	24 / 100 (Guarded)	Relevance guardrail correctly rejects candidate with 0% role match.
Junior Python Developer	Python Developer	68 / 100 (Uninformative)	68 / 100 (Actionable)	Identifies 57.1% match; highlights missing Docker & Celery competencies.
Senior AI Engineer	AI Engineer	88 / 100	91 / 100 (Strong Match)	Validates PyTorch, TensorFlow, NLP, and quantified impact metrics.
Image / Scanned Resume	Data Analyst	40 / 100 (Silent Fallback)	Explicit Red Notice Banner	Intercepts unparseable stream and prompts for valid PDF/DOCX text file.

3. Complete Modular Component & File Registry

MODULE / FILE PATH	ARCHITECTURAL LAYER	KEY TECHNICAL RESPONSIBILITY & ENGINEERING LOGIC	TECHNOLOGY STACK
core/parser.py	Extraction Layer	Multi-format PDF buffer and DOCX XML block parsing with UTF-8 normalization.	pypdf, python-docx
core/section_analyzer.py	Audit Layer	RFC email validation, phone regex, LinkedIn/GitHub URL detection, 5-section checklist.	Python re, Sanitizer
core/ats_scorer.py	Scoring Layer	Lookaround regex token matching, 50+ master skills taxonomy, guardrail enforcement.	Lookaround Regex, JSON
core/feedback_generator.py	Recommendation Layer	Google X-Y-Z formula bullet point generator and weak vs strong phrasing comparisons.	Rule-Based NLP
templates/components/	Presentation Layer	10 decoupled component templates (metrics, keyword badges, section checks, Google X-Y-Z).	Jinja2, HTML5, CSS Grid
app.py & Procfile	Gateway & DevOps	Flask routing, upload handling, security sanitization, and Gunicorn WSGI startup.	Flask 3.1, Gunicorn 23.0